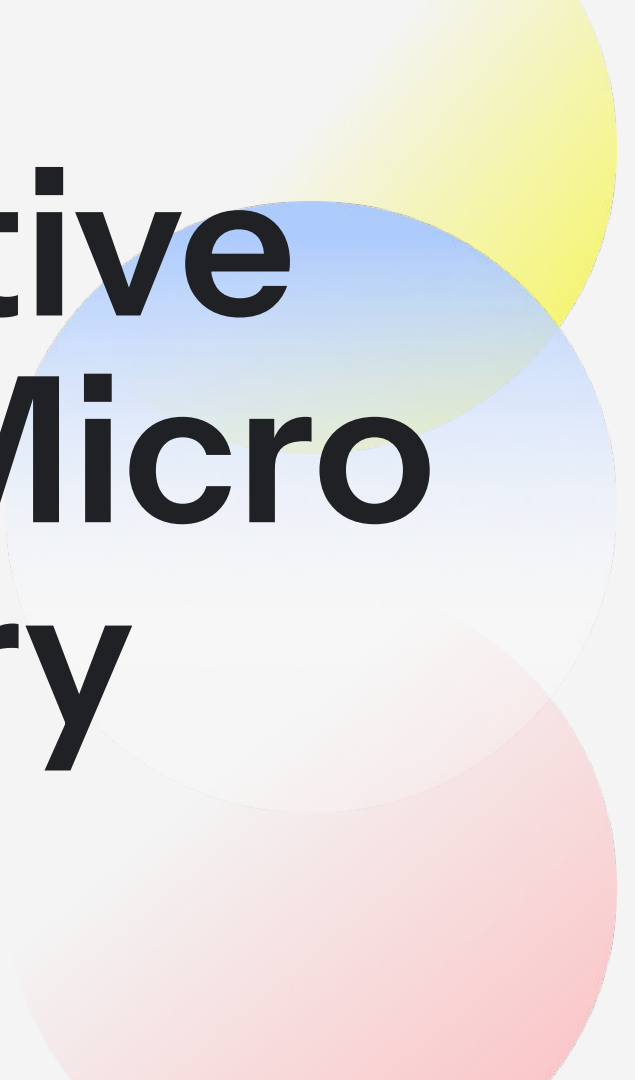


Generative Fashion Micro Factory



Diondria Morris

ITAI-3377

Use Case & AI Approach

- Portable booth produces on demand garments
- Edge appliance hosts three models:

**Diffusion
engine:
Text/Sketc
h $\rightarrow$
Pattern**

**Conversational
LLM: Collects
requirements for
garment**

**RL Scheduler:
Minimizes waste
& wait times**

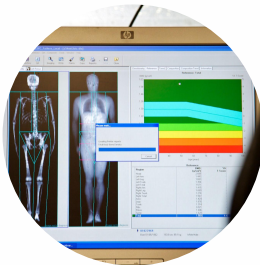
Key Architectural **Decisions**

- Rugged carry on with GPU or TPU
- MQTT V5 event bus (pub sub)
- OPC UA for cutter & sewing control
- RTSP stream for vision QC
- All inference local; cloud sync optional

Security Measures



- TPM anchored certificates, mutual TLS



- Encrypted body scans delete after 24 hours



- Signed models with hash verification

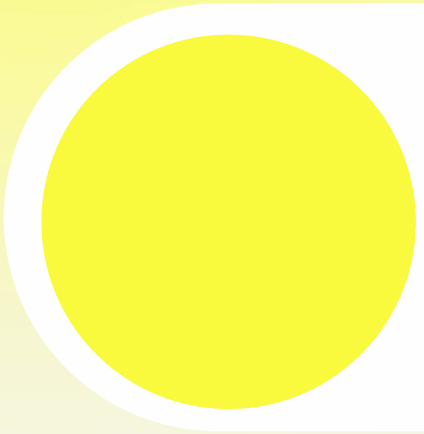


- Bias audits; cultural attribution warnings

Testing Strategy

- Functional Twin (SimPy): ≥ 95 % jobs ≤ 2 h
- Load Test (Locust): p95 ≤ 250 ms latency
- Red Team (Caldera): block unauthorized actions
- WAN Failover: production continues offlines
- Power Cycle: job resumes; $< 5\%$ fabric loss

Conclusion & Next Steps

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- Turns inspiration into clothes in 2 hours
 - Zero waste, zero cloud dependency
 - Prototype with simulated trials NEXT
 - Questions, Testing and Feedback